# UNCERTAINTY-AWARE RESPONSE ROUTING: WHEN DECOMPOSING UNCERTAINTY TYPES FALLS SHORT

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## ABSTRACT

Large language models increasingly need to calibrate responses based on confidence, yet current approaches typically reduce uncertainty to a single scalar for binary abstention decisions. We hypothesize that decomposing uncertainty into interpretable types—factual knowledge gaps, reasoning ambiguity, query underspecification, and temporal limitations—could enable more nuanced response routing. We propose Uncertainty-Aware Response Routing (UARR), training lightweight probes on LLM hidden states to classify uncertainty types. While our probes achieve 98.7% classification accuracy on constructed datasets, we find this success is largely driven by surface-level linguistic templates rather than genuine uncertainty signatures. Removing query modification templates drops accuracy from 98.7% to 74.3%, and errors concentrate between factual gaps and query underspecification—the distinction most critical for practical routing. These findings suggest that extracting reliable uncertainty type signals from LLM hidden states remains an open challenge.

## 1 INTRODUCTION

Large language models (LLMs) frequently produce confident-sounding outputs even when uncertain (Farquhar et al., 2024). Current approaches to handling uncertainty typically reduce it to a single scalar for binary abstention decisions (Wang et al., 2025; Chen & Varoquaux, 2025). This oversimplification ignores the rich structure underlying model uncertainty. We argue that different uncertainty types warrant different response strategies: a model uncertain due to lacking factual knowledge might benefit from r eval augmentation (Lewis et al., 2020), while an ambiguous query should trigger clarification requests.

We propose Uncertainty-Aware Response Routing (UARR), a framework that decomposes LLM uncertainty into four interpretable types and routes queries to appropriate response modes. Our approach trains lightweight probes (Belinkov, 2021) on hidden state representations to classify uncertainty types. Our experiments reveal a sobering finding: while UARR achieves 98.7% classification accuracy, this performance is largely attributable to surface-level linguistic cues rather than genuine uncertainty signatures. Ablation studies show that removing query modification templates drops accuracy to 74.3%, with errors concentrating between factual knowledge gaps and query underspecification—the very distinction most important for practical applications.

Our contributions are: (1) a taxonomy of uncertainty types with corresponding response strategies; (2) a probe-based classification framework achieving high accuracy on constructed data; and (3) ablation analysis revealing that high performance depends critically on template-based lexical cues, suggesting current probing approaches may be insufficient for robust uncertainty decomposition.

## 2 RELATED WORK

**Uncertainty Quantification in LLMs.** Recent work has made significant progress on detecting when LLMs are uncertain. Semantic entropy (Farquhar et al., 2024) computes uncertainty at the meaning level by clustering semantically equivalent generations, while semantic entropy probes (Kossen et al., 2024) approximate this using hidden states, avoiding expensive sampling. Token-level uncertainty (Fadeeva et al., 2024) and internal state analysis (Chen et al., 2024a) focus on hallucination detection but treat uncertainty as monolithic. SConU (Wang et al., 2025) applies conformal prediction for selective uncertainty with coverage guarantees. Query-level approaches (Chen & Varoquaux, 2025) aggregate token uncertainties but still produce single scalar outputs. All these methods use a single confidence measure for binary decisions, missing the opportunity for differentiated responses.

**Calibration and Probing.** Fine-tuning approaches (Chaudhry et al., 2024) teach models to express uncertainty linguistically, but require modifying model weights. Calibration methods (Bohdal et al., 2021; Chen et al., 2024b) improve confidence reliability but don't distinguish uncertainty sources. Probing classifiers (Belinkov, 2021) have been widely used to analyze what information neural representations encode. However, probing methodology has known limitations: high probe accuracy may reflect the probe's own learning capacity rather than information explicitly encoded in representations. Our findings reinforce this concern, showing probes can achieve high accuracy by exploiting superficial correlations.

## 3 Method

**Uncertainty Type Taxonomy.** We define four types based on the appropriate response strategy: (1) *Factual knowledge gaps*: the model lacks information needed to answer correctly—appropriate response is r eval augmentation or knowledge base lookup; (2) *Reasoning path ambiguity*: multiple valid reasoning approaches exist with potentially different conclusions—appropriate response is presenting multiple perspectives; (3) *Query underspecification*: the question itself is ambiguous or lacks necessary context—appropriate response is requesting clarification; (4) *Temporal limitations*: the answer depends on information beyond training cutoff—appropriate response is acknowledging temporal constraints or web search.

**Dataset Construction.** We construct an uncertainty-labeled dataset from three QA benchmarks: TriviaQA (Joshi et al., 2017), AmbigQA (Min et al., 2020), and Natural Questions (Kwiatkowski et al., 2019). To create distinguishable examples, we apply query modification templates: factual questions receive prefixes like "What is the exact…"; reasoning questions use "What are multiple ways to…"; underspecified queries have ambiguous pronouns inserted; temporal questions receive "As of today…". This yields approximately 2,000 examples with balanced class distribution (500 per type). While this construction enables controlled experiments, we acknowledge it introduces a confound between uncertainty type and surface-level linguistic patterns.

**Probe Architecture.** We extract hidden states from DistilGPT-2's last layer using mean pooling across token positions, then apply z-score normalization per feature dimension. The probe is a two-layer MLP (768→256→4) with batch normalization, GELU activations, and dropout (0.4). We include a learnable temperature parameter for calibration and use cross-entropy loss with label smoothing (0.1) and class-weighted sampling to handle any class imbalance.

## 4 Experiments

We evaluate on stratified train/validation/test splits (70%/15%/15%) using AdamW with OneCycleLR scheduling for 80 epochs with early stopping (patience 15). Training converges rapidly: both loss and accuracy plateau by epoch 10, with training and validation curves closely aligned throughout, suggesting the task may be easier than expected. The model achieves 98.7% test accuracy with Expected Calibration Error (ECE) of 0.090 and routing precision of 0.993. Per-class accuracies are 97.3% for factual gaps, 100% for reasoning ambiguity, 97.3% for query underspecification, and 100% for temporal limitations.

**Error Analysis.** Figure 1 reveals that all four classification errors occur between factual knowledge gaps and query underspecification (two in each direction). This symm c confusion pattern is particularly concerning: these two categories require opposite routing strategies—r eval augmentation versus clarification requests. A system that confuses "the model doesn't know the answer" with "the question itself is unclear" would either waste resources on unnecessary r eval or frustrate users with inappropriate clarification requests. The probe cannot reliably distinguish these cases even with template assistance.

**Critical Ablation: Query Templates.** Figure 2 presents our most important finding. Removing query modification templates drops accuracy from 98.7% to 74.3%—a 24.4 percentage point de-

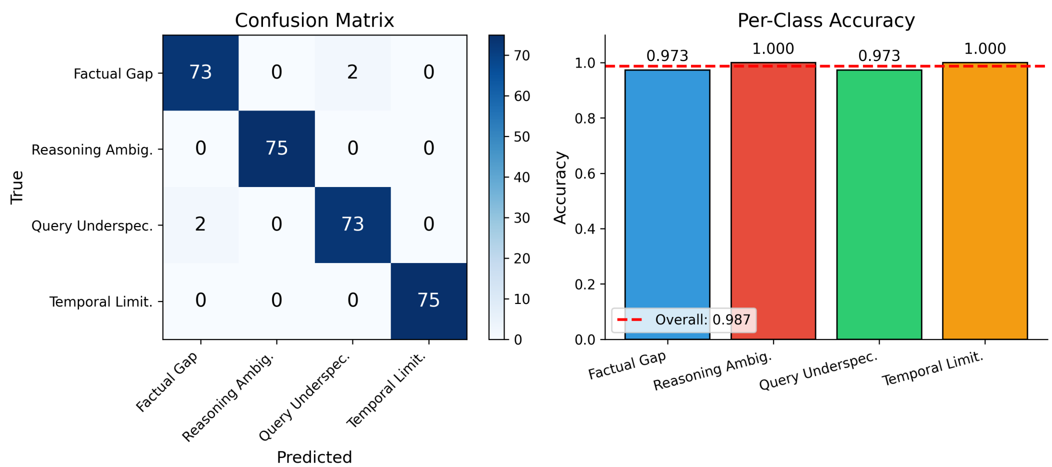


Figure 1: Left: Confusion matrix showing errors concentrate between factual gaps and query underspecification (2 errors each direction, 4 total). Right: Per-class accuracy with 98.7% overall. The balanced dataset contains 75 test samples per class.

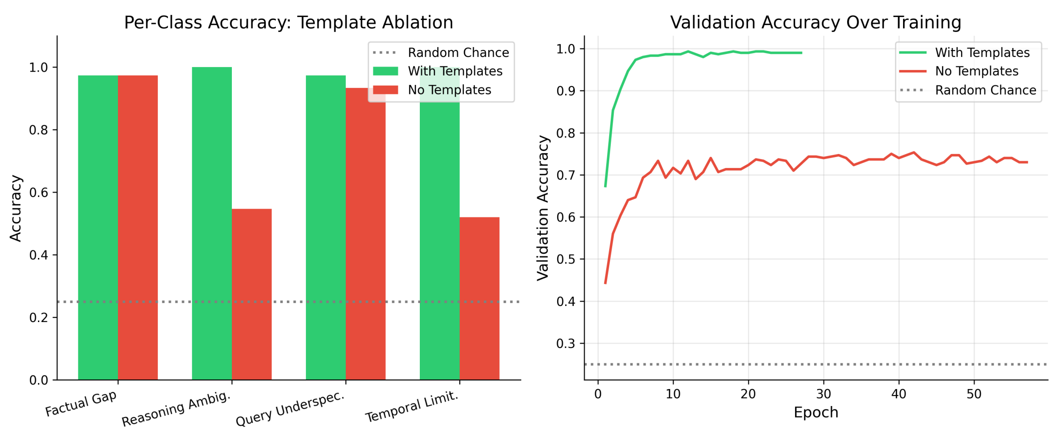


Figure 2: Template ablation results. Left: Per-class accuracy showing catastrophic drops for reasoning ambiguity (100%→54.7%) and temporal limitations (100%→52.0%) without templates. Right: Validation accuracy over training—with templates, the model converges to 99%; without, it plateaus at 74%.

crease. The per-class breakdown reveals highly asymm c damage: reasoning ambiguity drops from 100% to 54.7%, and temporal limitations from 100% to 52.0% (near random chance for 4 classes), while factual gaps (97.3%) and underspecification (93.3%) remain relatively stable. This pattern strongly suggests the probe learns template-specific lexical patterns (e.g., “multiple ways to” for reasoning, “As of today” for temporal) rather than genuine uncertainty signatures encoded in the hidden states. The categories that relied most heavily on distinctive templates collapse without them.

**TF-IDF Baseline.** To further investigate the role of lexical cues, we compare neural embeddings against TF-IDF features. TF-IDF achieves 92.7% accuracy—only 6 percentage points below the neural approach. This confirms that the majority of the classification signal is lexical rather than semantic. Interestingly, TF-IDF struggles most with factual gaps (78.7% accuracy), while achieving perfect accuracy on temporal limitations, likely due to highly distinctive temporal markers like “As of today” and “Currently in 2024.”

Table 1: Ablation study results. The template ablation reveals high accuracy depends heavily on surface-level linguistic cues rather than deep uncertainty representations in the hidden states.

| Configuration | Test Acc. | ECE | Routing Prec. |
|---|---|---|---|
| Full UARR | 0.987 | 0.090 | 0.993 |
| No Feature Normalization | 0.973 | 0.143 | 0.993 |
| TF-IDF Features | 0.927 | 0.120 | 0.970 |
| **No Templates** | **0.743** | 0.089 | 0.765 |

Table 1 summarizes additional ablation findings: feature normalization is critical for calibration (ECE increases from 0.090 to 0.143 without it), while architectural choices have minimal impact on accuracy. The template ablation stands out as the most impactful factor.

**Pooling Strategy.** We compared mean pooling versus last-token pooling for hidden state extraction. Mean pooling achieves 98.7% versus 95.7% for last-token, suggesting query-level information relevant to classification is distributed across positions. This contrasts with typical language modeling where the last token aggregates sequence information, and may reflect that template markers appearing at query beginnings contribute significant signal.

## 5 Discussion and Limitations

Our results highlight a fundamental challenge in uncertainty decomposition: the gap between what probes can classify and what LLMs actually encode. The 24.4% accuracy drop without templates suggests that DistilGPT-2's hidden states may not contain easily extractable uncertainty type information beyond surface lexical features. Several limitations apply: (1) we use DistilGPT-2 rather than larger models that might encode richer uncertainty representations; (2) our dataset construction methodology introduces template-uncertainty correlations by design; (3) we probe only the final layer, while uncertainty information might be distributed across layers.

The confusion between factual gaps and query underspecification is particularly problematic for real-world deployment. These categories represent the most common uncertainty types users encounter, yet require opposite interventions. A production system based on our approach would misroute approximately 2.7% of queries between these categories even with templates, and substantially more without them.

## 6 Conclusion

We proposed UARR for decomposing LLM uncertainty into interpretable types. While our probes achieve high classification accuracy, the template ablation demonstrates that without explicit lexical cues, accuracy drops by 24.4 percentage points. Furthermore, confusion between factual knowledge gaps and query underspecification persists even with templates. These findings suggest that extracting reliable uncertainty type signals from LLM internals is harder than simple probing approaches can achieve. Future work should explore whether larger models encode richer uncertainty representations, whether attention patterns provide complementary signals, and whether contrastive learning on naturally-occurring uncertainty examples (rather than template-constructed ones) yields more robust classifiers.

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# Supplementary Material

## A Dataset Construction Details

We construct uncertainty-labeled examples from three HuggingFace datasets. For **factual knowledge gaps**, we sample 500 questions from TriviaQA (Joshi et al., 2017) and optionally prepend factual markers ("What is the exact…", "Who specifically…", "When precisely did…", "Where exactly…"). For **reasoning path ambiguity**, we transform 500 Natural Questions (Kwiatkowski et al., 2019) using reasoning templates ("What are multiple ways to…", "Compare different approaches to…", "What factors influence…", "Analyze the tradeoffs in…"). For **query underspecification**, we use 500 AmbigQA (Min et al., 2020) questions, inserting ambiguous markers ("it", "they", "that thing", "the person", "somewhere"). For **temporal limitations**, we prepend temporal markers ("As of today,", "Currently in 2024,", "What is the latest…", "Recent updates on…", "Breaking news about…") to 500 Natural Questions.

# B Training Hyperparameters

We use the following hyperparameters: hidden dimension 256, dropout 0.4, learning rate 0.001 with OneCycleLR scheduling, weight decay 0.05, batch size 32, maximum epochs 80 with early stopping patience 15, label smoothing 0.1, and class-weighted cross-entropy loss. Hidden states are extracted from DistilGPT-2's last layer using mean pooling over token positions, followed by z-score normalization.

# C Training Dynamics

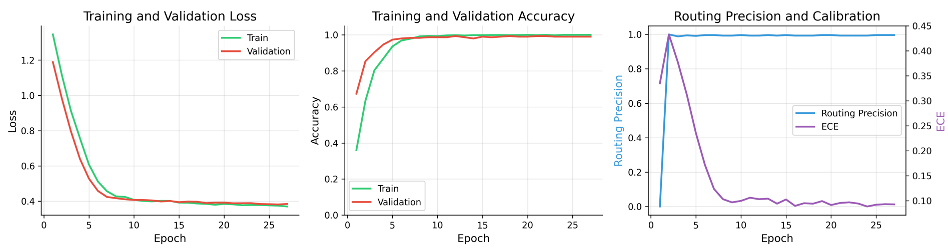


Figure 3: Training dynamics showing rapid convergence. Loss and accuracy plateau by epoch 10, routing precision saturates near 1.0 by epoch 4, and ECE stabilizes around 0.09. Early stopping triggered around epoch 25. The close alignment between training and validation curves suggests the classification task presents limited generalization challenge.

Figure 3 shows training dynamics across epochs. The rapid convergence (plateau by epoch 10) and minimal train-validation gap suggest that the task, as constructed with templates, does not pose a significant learning challenge for the probe architecture.

# D TF-IDF Baseline Analysis

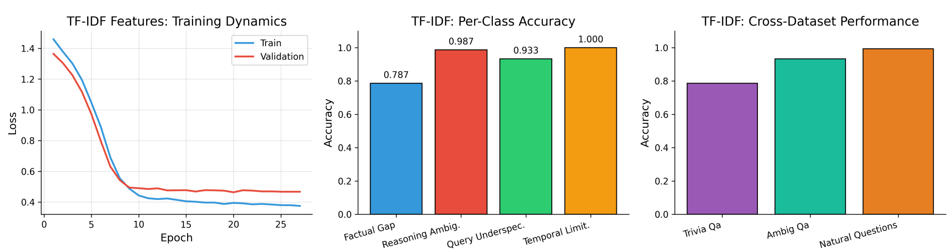


Figure 4: TF-IDF baseline results. Left: Training dynamics show convergence with a larger train-validation gap compared to neural embeddings. Middle: Per-class accuracy reveals factual gaps are hardest (78.7%) while temporal limitations achieve 100%. Right: Cross-dataset performance showing variation across source datasets.

Figure 4 analyzes the TF-IDF baseline. The 92.7% overall accuracy achieved by purely lexical features confirms that template-based construction introduces strong surface-level signals. The per-class breakdown is informative: temporal limitations achieve perfect accuracy due to highly distinctive markers ("As of today", "Currently in 2024"), while factual gaps are most challenging (78.7%), suggesting these questions have more varied linguistic structure.

# E Pooling Strategy Comparison

Figure 5 compares pooling strategies. Mean pooling outperforms last-token pooling by 3 percentage points, contrasting with typical language modeling where the last token aggregates sequence

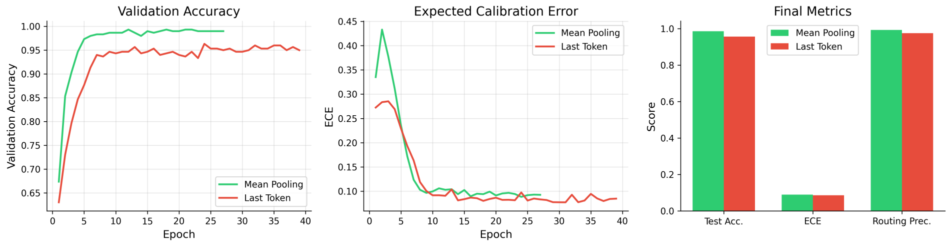


Figure 5: Comparison of mean pooling vs. last-token pooling. Mean pooling achieves higher accuracy (98.7% vs. 95.7%) and faster convergence. This suggests query-level information relevant to uncertainty classification is distributed across token positions rather than concentrated at the sequence end.

information. This may indicate that uncertainty-relevant features (particularly template markers that appear at query beginnings) are distributed throughout the hidden state sequence.

# F Cross-Dataset Generalization

Performance across source datasets is uniformly high: TriviaQA (97.3%), AmbigQA (97.3%), and Natural Questions (100%). However, this may partially reflect dataset identification rather than genuine uncertainty classification, since uncertainty type labels correlate with source dataset by construction. Natural Questions achieves perfect accuracy as it provides examples for both reasoning ambiguity and temporal limitations—the two categories most dependent on template modifications.